Optimization Using Numerical Methods

Electricity Bill Minimisation via Linear Programming

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Outline

1. Introduction
2. Data & Tariff
3. LP Formulation
4. KKT Analysis
5. Results
6. Conclusion

Research Problem

- Household electricity bills depend on **when** energy is used.
- Peak hours (18:00–22:00) $\rightarrow$ higher tariff; off-peak (23:00–06:00) $\rightarrow$ lower tariff.
- Time-flexible appliances (fan, TV, motor pump) can be rescheduled to off-peak windows.
- Goal: Re-allocate usage to minimise the annual bill without reducing service.

Research Question

"By how much can the annual electricity bill be reduced using optimal time-of-use scheduling, and what is the per-unit savings efficiency?"

Dataset

- Kaggle dataset: 45,345 household-month observations.
- Six appliances: Fan, Refrigerator, AirConditioner, Television, Monitor, MotorPump.
- Variables: monthly energy (kWh), tariff rate (Rs./kWh), bill (Rs.), month.
- MotorPump column is zero in all records.
- Mean tariff $\bar{c} = 8.3696$ Rs./kWh, SD $\sigma = 0.5770$.

Tariff Construction

Peak / Off-peak rates

$$c^{(p)} = 1.2\,\bar{c} = 10.0436 \text{ Rs./kWh}$$

$$c^{(o)} = 0.8\,\bar{c} = 6.6957 \text{ Rs./kWh}$$

- Ratio $c^{(p)}/c^{(o)} = 1.5$ (the only parameter driving savings).
- All demand originally assumed to be charged at peak rate.

Decision Variables & Parameters

R_i : total annual energy requirement of appliance i (kWh)

x_i : energy allocated to the off-peak schedule (kWh)

$n = 6$ appliances

Total system demand: $D = \sum_{i=1}^{n} R_i = 23,83,687$ kWh.

Linear Programme

$$\min_{\mathbf{x}} \; Z(\mathbf{x}) = \sum_{i=1}^{n} c^{(o)} x_i = 0.8\,\bar{c} \sum_{i=1}^{n} x_i$$

subject to

$$\begin{cases} x_i \geq R_i, & i = 1, \ldots, n \quad \text{(demand satisfaction)} \\ x_i \geq 0, & i = 1, \ldots, n \quad \text{(non-negativity)} \end{cases}$$

- The objective is linear, the feasible set is a convex polyhedron.
- $\Rightarrow$ convex programme $\rightarrow$ any local minimum is global.

Lagrangian & KKT Conditions

Lagrangian (with multipliers $\lambda_i \geq 0$ for $x_i \geq R_i$, $\mu_i \geq 0$ for $x_i \geq 0$):

$$\mathcal{L} = c^{(o)} \sum_i x_i - \sum_i \lambda_i (x_i - R_i) - \sum_i \mu_i x_i$$

$$\begin{aligned}
\text{Stationarity:} \quad & c^{(o)} - \lambda_i - \mu_i = 0 \quad \forall i \\
\text{Complementarity:} \quad & \lambda_i (x_i - R_i) = 0, \quad \mu_i x_i = 0 \\
\text{Primal feasibility:} \quad & x_i \geq R_i, \; x_i \geq 0 \\
\text{Dual feasibility:} \quad & \lambda_i, \mu_i \geq 0
\end{aligned}$$

Unique Optimal Solution

- Because $c^{(o)} > 0$, we must have $\mu_i = 0$ whenever $x_i > 0$.
- Then $\lambda_i = c^{(o)} > 0 \Rightarrow$ binding constraint: $x_i^* = R_i$.
- All demand is shifted to the off-peak window.
- The optimal cost is

$$Z^* = 0.8\,\bar{c}\sum_i R_i = 0.8\,\bar{c}\,D = \text{Rs. } 1,59,60,497.$$

Why Exactly $\frac{1}{3}$ Savings?

$$\frac{Z_{\text{before}}}{Z^*} = \frac{1.2\,\bar{c}\,D}{0.8\,\bar{c}\,D} = \frac{1.2}{0.8} = \frac{3}{2}$$

$$\Rightarrow Z^* = \frac{2}{3}\,Z_{\text{before}}, \qquad \text{Saving} = \frac{1}{3}\,Z_{\text{before}} \approx 33.3\%.$$

The saving depends only on the peak/off-peak ratio, not on the actual tariff level.

Appliance-Level Savings

Appliance	Demand (kWh)	Before (Lakh)	After (Lakh)
Fan	6,34,408	63.72	42.48
Refrigerator	9,84,234	98.85	65.90
Television	5,66,932	56.94	37.96
AirConditioner	68,197	6.85	4.57
Monitor	1,29,916	13.05	8.70
MotorPump	0	0	0
Total	**23,83,687**	**239.41**	**159.60**

Total Saving = 79.80 Lakh (33.3%), Efficiency = 3.35/kWh

Stochastic Robustness

Tariff $C \sim \mathcal{N}(\mu, \sigma^2)$ with $\mu = 8.3696$, $\sigma = 0.5770$.

$$Z_{\text{before}} = 1.2\, C\, D, \qquad Z^* = 0.8\, C\, D$$

$$\mathbb{E}[Z_{\text{before}}] = 1.2\, \mu\, D, \qquad \mathbb{E}[Z^*] = 0.8\, \mu\, D$$

- The saving $S = 0.4\, C\, D$ is positive with probability 1 (since $C_{\min} = 7.40 > 0$).
- The LP optimal solution remains optimal in expectation.
- Standard deviations: $\text{SD}[Z_{\text{before}}] = 16.50$ Lakh, $\text{SD}[Z^*] = 11.00$ Lakh.

Summary

- All six appliances can be served entirely at the off-peak tariff under the idealised model.
- The optimised bill is exactly 2/3 of the original, giving 33.3% savings.
- Savings are scale-invariant: they stem only from the tariff ratio.
- The result is robust to normally distributed tariff uncertainty.
- Annual saving: **Rs. 79,80,249** (3.35 per kWh shifted).

Limitations

- The two-period simplification ignores hourly tariff variations.
- Refrigerator (41% of demand) cannot be fully shifted; a flexibility cap $x_i \leq \alpha_i R_i$ would reduce savings.
- MotorPump is zero in the dataset $\rightarrow$ no estimate of its shift potential.
- Data are aggregated over 45,345 households, not a single-home longitudinal panel.

Future Work

1. Hourly-resolved LP with true time-of-day tariffs.
2. Incorporate comfort constraints via quadratic penalty (QP).
3. Use smart-meter data for personalised, per-household optimisation.
4. Stochastic programming with explicit tariff distribution.

Any Questions?

Thank you!